MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, SEVEN TWISTED PAIRS NO. 16 GAUGE FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 7PR-16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.
Second - Insulation of ethylene tetrafluoroethylene (TEFZEL). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.
Third - Two insulated conductors, one black and one white, twisted together with a nominal lay of 2-1/2 inches to form each pair.
Fourth - Marker braid on each pair, standard identification code applied by method 4.
Fifth - Seven pairs cabled with a lay not greater than 24 times the pitch diameter of the layer.
Sixth - Rubber filler in order to form a firm, round assembly. Filler shall be capable of removal from the assembly without damaging the insulation on conductors.
Seventh - A binder tape of polyester, applied helically with overlap.
Eighth - Cable jacket of black polyether polyurethane. The cable jacket material shall be B.F. Goodrich Co. ESTANE 58300 or 58863; or Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.100 inch. Overall cable diameter to be 0.630 to 0.650 inch.

FSC 6145
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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - at 20°C, ohms/1000 feet, maximum.	4.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to water.....	2000
Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Insulation resistance - megohms/1000 feet, minimum.....	5000
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - at minus 40 ± 2°C over a 4-inch diameter mandrel.....	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	5000
Elongation - percent, minimum.....	100
Cable jacket	
Tensile strength - lbf/in^2, minimum.....	5000
Elongation - percent, minimum.....	500
Group C:	
Physicals (aged)	
Insulation	
Air oven (168 hours at 180°C)	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	90
Cable jacket	
Air oven (168 hours at 80°C)	
This test shall be in accordance with MIL-C-915.....	No failure
Hot oil immersion	
Tensile strength, percent of unaged, minimum.....	90
Elongation, percent of unaged, minimum...	80

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QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0886-03)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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